Sequences and series CHEAT SHEET

SEQUENCES

$a_1, a_2, a_3, \ldots, a_n$ a_n = the nth term

$a_n = 2n + 1$: 3, 5, 7, 9, ... $a_{100} = 201$

$a_n = 3n - 2 = 100 \rightarrow n = 34$

5, 8, 11, 14, ... $\rightarrow$ $a_n = 3n + 2$

recursive: $a_1 = 3$, $a_n = a_{n-1} + 4$: 3, 7, 11, 15

Fibonacci: $a_n = a_{n-1} + a_{n-2}$: 1, 1, 2, 3, 5, 8, 13, ...

$\frac{55}{34} \approx 1.618 = \frac{1+\sqrt{5}}{2}$ (the golden ratio)

circle regions 1, 2, 4, 8, 16, then 31: a few terms don't fix a rule

SERIES AND SIGMA NOTATION

$$S_n = a_1 + a_2 + \ldots + a_n = \sum_{k=1}^{n} a_k$$

$$\sum_{k=1}^{5} k^2 = 1 + 4 + 9 + 16 + 25 = 55$$

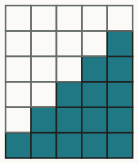
$$\sum_{k=3}^{6} k = 3 + 4 + 5 + 6 = 18$$

$$\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$$

$$\sum_{k=1}^{n} c\,a_k = c \sum_{k=1}^{n} a_k \qquad \sum_{k=1}^{n} 1 = n$$

$$\sum_{k=1}^{10} (3k + 2) = 3 \cdot 55 + 2 \cdot 10 = 185$$

ARITHMETIC PROGRESSIONS (AP)



two copies of 1 + 2 + ... + 5
make a 5 × 6 rectangle:
1 + ... + 5 = 30 ÷ 2 = 15

common difference $d = a_{n+1} - a_n$

$a_n = a + (n-1)d$ (10 posts, 9 gaps)

$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

$1 + 2 + \ldots + 100 = 50 \times 101 = 5050$

$1 + 3 + 5 + \ldots + (2n-1) = n^2$

arithmetic mean $A = \frac{a+b}{2}$ 8, 14, 20

n means between a and b: $d = \frac{b-a}{n+1}$

WORKED AP EXAMPLES

jar: 50, 70, 90, ... week 30: $50 + 29 \cdot 20 = 630$

30 weeks: $\frac{30}{2}(50 + 630) = 10\,200$ coins

24, 21, 18, ...: 78 after 4 terms and after 13

theatre: 20 seats, +2 a row, 25 rows: last 68, total 1100

two-digit multiples of 3: 30 terms, sum 1665

GEOMETRIC PROGRESSIONS (GP)

common ratio $r = a_{n+1} \div a_n$ (no term 0)

$a_n = ar^{n-1}$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$ $r = 1$: $S_n = na$

why: $S - rS = a - ar^n$ (the shifted row cancels)

2, 8, 32, ...: $2 \cdot 4^{n-1} = 131\,072 \rightarrow n = 9$

x% growth $= \times(1 + \frac{x}{100})$: $1000 \times 1.05^{10} \approx 1629$

chessboard: $1 + 2 + \ldots + 2^{63} = 2^{64} - 1 \approx 1.8 \times 10^{19}$

THE SUM TO INFINITY

each piece fills half the gap left: the total closes in on 2

$-1 < r < 1$: $S_\infty = \frac{a}{1-r}$ otherwise no sum to infinity

$1 + \frac{1}{2} + \frac{1}{4} + \ldots = 2$ ball: $2 + 2 \times 2 = 6$ m

$0.333\ldots = \frac{3/10}{1 - 1/10} = \frac{1}{3}$ $0.2727\ldots = \frac{27}{99} = \frac{3}{11}$

$0.999\ldots = \frac{9/10}{9/10} = 1$

swing: 3 m, then 80% each time: $\frac{3}{1-0.8} = 15$ m

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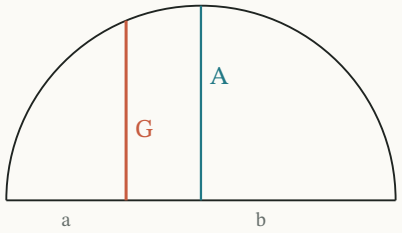
THE ARITHMETIC AND GEOMETRIC MEANS



radius = (a + b)/2 = A
half-chord: $h^2 = ab$, h = G
no half-chord beats
the radius: A ≥ G

geometric mean $G = \sqrt{ab}$ $(a, b > 0)$ 4, 6, 9

growth × 4 then × 9: steady × 6, not × 6.5

n means between a and b: $r = (\frac{b}{a})^{1/(n+1)}$

1, 4, 16, 64, 256 3, 9, 27, 81

$A - G = \frac{(\sqrt{a} - \sqrt{b})^2}{2} \geq 0$, so $A \geq G$

equal only when a = b 4 and 9: $6.5 - 6 = \frac{1}{2}$

A = 10, G = 8 → a + b = 20, ab = 64 → 16 and 4

40 m of fence: area ≤ $10^2 = 100$ m² (the square)

MONEY AND GROWTH

simple interest (an AP): 1000, 1060, 1120, … 10 years: 1600

compound (a GP): 1000×1.06^n 10 years ≈ 1791, 30 years ≈ 5743

save D at each year's end: $S = D\,\frac{(1+i)^n - 1}{i}$

1000 a year, 5 years, 5%: ≈ 5526 coins

100 a month, 12 months, 1% a month: ≈ 1268 coins

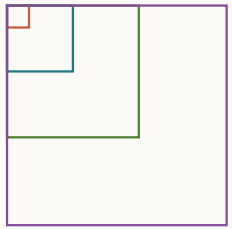
loan L, n payments P: $L(1+i)^n = P[(1+i)^{n-1} + \ldots + 1]$

33 100 at 10%, 3 years: $P = \frac{44\,056.1}{3.31} = 13\,310$

town 50 000, +2% a year: $1.02^{10} \to \approx 60\,950$

year	owed	+ 10%	− 13 310 →
1	33 100	3 310	23 100
2	23 100	2 310	12 100
3	12 100	1 210	0

SUMS OF SPECIAL SERIES



$1^3 + 2^3 + 3^3 + 4^3$
fill a square of side
1 + 2 + 3 + 4 = 10: 100

$1 + 2 + \ldots + n = \frac{n(n+1)}{2}$ $1 + \ldots + 20 = 210$

$1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$ to 10: 385

$1^3 + 2^3 + \ldots + n^3 = [\frac{n(n+1)}{2}]^2$ to 10: $55^2 = 3025$

proof of the squares: $(k+1)^3 - k^3 = 3k^2 + 3k + 1$, added up

cubes: 1 | 3 + 5 | 7 + 9 + 11 = 1, 8, 27 (odd numbers)

SERIES BUILT FROM THEM

$1 \cdot 2 + 2 \cdot 3 + \ldots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

$1 + 3 + 6 + 10 + \ldots = \frac{n(n+1)(n+2)}{6}$ 10 layers: 220

$5^2 + \ldots + 20^2 = 2870 - 30 = 2840$

$3 \cdot 8 + 6 \cdot 11 + 9 \cdot 14 + \ldots = 3n(n+1)(n+3)$

recipe: write the kth term, split it, use the three sums